Grain Definition Document

AeroGrid

Low-Altitude Airspace Intelligence Warehouse

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1 Introduction

Defining precise grain statements for each fact table is critical to prevent double-counting, ensure data integrity, and guide physical database implementation in the AeroGrid Data Warehouse.

2 Fact Table Grain Statements

1. **FactVehicleTracking Grain:**
   "One row represents one position report (state vector) from one vehicle at one exact timestamp."
   Purpose: Enables instantaneous density calculations and continuous position tracking.

2. **FactFlightJourney Grain:**
   "One row represents a complete flight journey or distinct flight segment from takeoff or monitored area entry to landing or exit."
   Purpose: Evaluates route choice efficiency, journey duration, and altitude profiles.

3. **FactCloseEncounter Grain:**
   "One row represents a detected safety conflict event where two vehicles breach minimum safe horizontal and vertical separation thresholds."
   Purpose: Quantifies near-midair collision risks across airspace sectors and weather conditions.

4. **FactRestrictionActivation Grain:**
   "One row represents an active restriction event or TFR evaluation per time interval."

Purpose: Measures regulatory compliance and traffic rerouting around restricted airspace.

5. **FactAirspaceOccupancy Grain:**
   "One row represents a periodic (hourly) utilization snapshot of vehicles occupying a specific spatial grid cell and altitude band."
   Purpose: Supports long-term capacity planning and corridor congestion analysis.